

Project Voxera: Background-Invariant, Egocentric Indian Sign Language Recognition via ResNet18 2D-CNN Inference Deployed on a Wearable Edge Device

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Abstract—Project Voxera is a wearable, egocentric sign language recognition system designed to eliminate the structural communication barriers experienced by the 430 million-strong global community of Deaf and hard-of-hearing (DHH) individuals. The system is motivated by the *Interpretation Dependency* problem: the condition by which a DHH individual’s ability to participate in any communicative event is contingent on the physical presence of a qualified interpreter—a dependency that imposes an estimated USD 20 800 per year in interpreter costs on the average DHH professional and contributes to a 23–38% earnings gap relative to hearing peers [1]. Unlike prior skeleton-based architectures—including the authors’ own discontinued MediaPipe Holistic LSTM prototype, which failed to meet the 33 ms real-time frame budget on the ARM Cortex-A53 target processor even after INT8 quantization (47 ms FP32, 38 ms INT8) and suffered a 4.2% accuracy degradation from LSTM gate quantization noise—the current system processes raw pixel frames of Indian Sign Language (ISL) static signs directly through a quantized ResNet18 convolutional neural network, requiring no third-party skeletal landmark tracker at inference time. The complete system pipeline comprises: (1) an original Point-of-View (POV) dataset of 40 ISL sign classes recorded using the chest-mounted OmniVision OV5647 fisheye camera of the deployed wearable device, eliminating domain shift between training and inference by construction; (2) an aggressive offline image augmentation scheme employing randomized brightness shifting, Gaussian noise injection, 2D affine rotation, horizontal flipping, random crop-and-resize, and colour jitter—expanding the 12 000-image raw corpus $8\times$ to approximately 96 000 training images; (3) a fine-tuned ResNet18 backbone with a custom 256-unit classification head achieving 92.3% top-1 accuracy on the 40-class ISL vocabulary in FP32 precision; and (4) INT8 post-training quantization via TensorFlow Lite, compressing the model from 44.7 MB to 11.2 MB ($4.0\times$ ratio) and reducing per-frame inference latency from 68 ms to 25 ms on the Radxa Zero 3W single-board computer (ARM Cortex-A53 quad-core @ 1.8 GHz), within the 33 ms frame budget for sustained 30 fps operation. Accuracy cost of quantization is 0.5 percentage points ($92.3\% \rightarrow 91.8\%$), within the 95% confidence interval of the evaluation ($\pm 0.9\%$). End-to-end latency from sign presentation to spoken-word audio output is $87\text{ ms} \pm 12\text{ ms}$ (median $\pm$ SD), well below the 150 ms perceptual threshold at which conversational delay becomes noticeable. Total hardware Bill of Materials cost is approximately USD 85. The system directly addresses UN SDG 4 (Quality Education) and SDG 8 (Decent Work and Economic Growth) by enabling cloud-free, interpreter-independent, real-time sign-to-speech communication from a wearable form factor.

Index Terms—Sign Language Recognition, Indian Sign Language (ISL), Egocentric Vision, ResNet18, Transfer Learning, Residual Networks, Data Augmentation, Edge Computing, Assistive Technology, TensorFlow Lite, INT8 Quantization, Wearable Computing, ARM Cortex-A53, Radxa Zero 3W, Sustainable Development Goals

I. INTRODUCTION AND THE SOCIOLOGICAL PROBLEM

A. The Global Communication Gap

There are an estimated 70 million Deaf individuals worldwide, with a user population of sign language as the primary mode of linguistic expression exceeding 430 million when accounting for the broader hard-of-hearing and sign-dependent population [1]. Sign languages are complete, grammatically complex, fully expressive natural languages that developed independently within Deaf communities across different national and cultural contexts over centuries of use. Indian Sign Language (ISL) is a fully autonomous visual-gestural language bearing no systematic grammatical relationship to Hindi, English, or any other spoken language of the Indian subcontinent [1]. Despite the full linguistic status of sign languages, the overwhelming architecture of modern society—its educational institutions, workplaces, legal systems, healthcare environments, and commercial infrastructure—is designed around the assumption of spoken-language fluency, creating a structural and systemic exclusion of DHH individuals from full participation in every domain of civic and professional life.

Studies conducted across the European Union between 2022 and 2024 found that Deaf adults earn, on average, between 23% and 38% less than hearing peers of equivalent educational attainment—a wage gap driven not by differences in cognitive capability or professional skill, but by communication friction at every stage of the professional pipeline [1]. In India, fewer than 300 certified ISL interpreters serve a DHH population of over 18 million, representing a demand-supply gap exceeding 99%.

B. The Interpretation Dependency Problem

The conventional accommodation mechanism for DHH individuals in hearing-centric environments is the provision of a qualified sign language interpreter. While effective, interpreter-mediated communication introduces a cascade of structural problems. Interpreters are expensive: professional rates in developed markets range from USD 60 to USD 150 per hour. The Registry of Interpreters for the Deaf reported a nationwide shortage exceeding 30% of projected demand as of 2024 [1]. More fundamentally, interpreter-mediated communication is inherently triadic—rather than a direct dyadic exchange, it requires a third party as communicative intermediary, disrupting eye contact and eliminating conversational spontaneity.

Formal Definition: The Interpretation Dependency. The Interpretation Dependency is the structural condition by which a DHH individual’s ability to participate in any communicative event is contingent upon the physical presence or remote availability of a qualified human interpreter or designated communication aide. This dependency imposes an estimated USD 20 800 per year in interpreter costs on a Deaf professional requiring interpretation for just 4 hours per work week [1].

C. The Stationary Device Dependency in Existing Technological Solutions

Over the past decade, a significant body of academic research and commercial development has been directed toward automating sign language recognition through computer vision and machine learning. However, without exception, existing systems introduce a new form of dependency in place of the Interpretation Dependency: the *Stationary Device Dependency*. Every existing sign language recognition application requires the signing hand to be presented to a camera placed on a tripod, held by a second person, or propped against a fixed surface. Furthermore, exocentric (third-person) camera systems suffer from well-documented angular variance problems [3]. A hand sign recognized at 90% accuracy from directly in front of the signer degrades to 61% accuracy when the camera is offset by just 45 degrees azimuth.

D. The Egocentric Vision Paradigm and Project Voxera

The Egocentric Vision paradigm resolves both the Stationary Device Dependency and the angular variance problem through a fundamental inversion of the observation axis. Instead of observing the signer from the outside—the exocentric perspective—an egocentric system observes the act of signing from the signer’s own chest-level or head-level perspective, via a camera mounted on the signer’s body. Research formalized by the Ego-Vision Consortium [4] demonstrated that egocentric training data reduces within-class spatial feature variance by a factor of 2.3 compared to exocentric training.

Project Voxera embodies this paradigm in its most practical form: a chest-mounted fisheye camera connected to a wearable single-board computer, forming a self-contained device that requires no environmental setup, no internet connection, and

no secondary device. The current implementation—Voxera Brain 3—targets ISL static signs using a ResNet18 convolutional neural network processing raw camera frames directly, achieving real-time performance on a USD 85 embedded device.

The remainder of this paper is organized as follows. Section II reviews the fundamentals of sign language processing. Section III surveys related work. Section IV documents the discontinued MediaPipe-LSTM prototype. Section V describes the data engineering pipeline. Section VI details the ResNet18 architecture and training methodology. Section VII covers hardware deployment. Section VIII presents experimental results and ablation study. Section IX analyses SDG impact. Section X concludes the paper.

II. FUNDAMENTALS OF SIGN LANGUAGE PROCESSING

A. The Phonological Parameters of ISL: HOLME

Signed languages are not mime, gesture, or pantomime. They are structured linguistic systems governed by a well-described phonological hierarchy. In ISL—as in ASL and other sign languages—five primary phonological parameters distinguish one sign from another, collectively described by the acronym HOLME: **H**andshape, **O**rientation, **L**ocation, **M**ovement, and **E**xpression. For the purposes of Voxera Brain 3, the most directly relevant parameters are Handshape and Orientation, as the system targets static neutral-space signs fully defined by a single held pose.

B. Static Anchor Signs: The Architectural Rationale for Single-Frame Classification

Voxera Brain 3 targets static ISL signs exclusively: the 26-letter manual alphabet and 14 common static-space vocabulary words. This restriction yields three decisive architectural advantages. First, static sign classification does not require temporal sequence modelling—a single frame is sufficient for high-confidence classification, eliminating the need for LSTMs, GRUs, or Transformers. Second, single-frame classification reduces the inference stack to a single forward pass through the ResNet18 network per camera frame, enabling the 25 ms INT8 latency achieved on the Cortex-A53. Third, a large static-sign image dataset is constructable from short video clips without temporal alignment annotation.

C. Vision-Based vs. Sensor-Based Sign Recognition

Two families of sign language recognition systems dominate the literature. Sensor-based systems instrument the hand with flex sensors, inertial measurement units, and electromyography arrays, achieving extremely high skeletal tracking accuracy (~99% landmark localization within 2 mm) but are socially invasive. Vision-based systems extract hand information from ordinary camera images, requiring no body instrumentation. Voxera adopts the vision-based approach for three reasons: non-invasiveness; compatibility with the chest-mounted wearable form factor; and the suitability of 2D CNN architectures for direct image classification without an intermediate skeletal extraction stage [3].

III. RELATED WORK

A. Skeleton-Based Sign Language Recognition

Skeleton-based sign language recognition systems extract 2D or 3D skeletal joint coordinates from video frames and feed these coordinate sequences into temporal classifiers. Das [3] provides a comprehensive analysis of architecture trade-offs in MediaPipe-based recognition systems for edge deployment, establishing that LSTM classifiers achieve high accuracy on moderate-size vocabularies (10–100 signs) but impose latency penalties on embedded ARM processors that can preclude real-time operation, reporting LSTM latencies of 41–53 ms on Cortex-A53 class processors—consistent with the authors’ own measurements of 47 ms FP32 / 38 ms INT8 in the discontinued prototype.

Graph Convolutional Networks (GCNs) operating on body skeleton graphs have demonstrated state-of-the-art accuracy on large-vocabulary sign recognition benchmarks, but their computational complexity is substantially higher than LSTM-based approaches, making them unsuitable for embedded deployment without aggressive model compression.

B. CNN-Based Direct Image Classification for Sign Recognition

An alternative and increasingly popular approach bypasses skeletal extraction entirely, training convolutional neural networks to classify hand signs directly from image pixels. He et al. [2] demonstrated that ResNet architectures, through their residual skip connections, are particularly well-suited to fine-tuning on small domain-specific datasets. Recent work has demonstrated that aggressive data augmentation—particularly combinations of colour jitter, random erasing, Gaussian noise, and synthetic background injection—significantly improves background invariance in CNN-based sign classifiers without architectural modifications.

C. Egocentric Vision for Gesture and Action Recognition

The Ego-Vision Consortium [4] established the theoretical and empirical basis for the egocentric paradigm in sign language recognition, demonstrating that egocentric training data reduces within-class spatial feature variance by a factor of 2.3 compared to exocentric training, due to the consistent spatial relationship between the signer’s hands and the chest-mounted camera. The Consortium also documented that a classifier trained on exocentric data and evaluated on egocentric data experienced an average accuracy drop of 18.4 percentage points.

D. Edge Deployment of Deep Learning Models

Key findings relevant to this work include: INT8 post-training quantization introduces negligible accuracy loss (<1%) for standard feedforward CNNs but can introduce significant degradation (>3%) for recurrent networks (LSTM, GRU) due to the sensitivity of sigmoid and tanh gate activations to quantization noise; INT8 quantization achieves approximately 4× model size reduction and 2–4× inference

speedup on ARM Cortex-A series processors with NEON SIMD support.

E. Research Gap Addressed by This Work

The existing literature identifies four gaps that Project Voxera directly addresses: (1) no prior ISL recognition system has been designed from the ground up for egocentric POV operation, eliminating domain shift by construction; (2) no prior work has provided a rigorous empirical comparison, on identical target hardware, of skeleton-based LSTM and CNN-based direct classification architectures; (3) the failure modes of MediaPipe Holistic on non-standard skeleton datasets have not been systematically documented; (4) no prior work has demonstrated a complete sign-to-speech pipeline on a self-contained wearable device costing under USD 100 with real-time operation and >90% accuracy.

IV. THE DISCONTINUED MEDIAPIPE-LSTM PROTOTYPE: AN ENGINEERING POST-MORTEM

A. Original Design

The original Voxera prototype proposed a six-stage pipeline: (1) WLASL dataset acquisition; (2) MediaPipe Holistic skeletal extraction producing 3D landmark coordinate sequences; (3) geometric filtering to reject non-neutral-space signs; (4) spatial and temporal normalization; (5) LSTM classification of (30,126) coordinate matrices to sign labels; and (6) INT8 TFLite quantization and Radxa Zero 3W deployment.

B. Failure Point 1: The JSON Parser Trap and Dataset Anomalies

The WLASL dataset’s JSON metadata file was expected to contain explicit video identification keys. No such key existed; the unique numeric video identifier was embedded within the file-path string of each entry. Standard JSON field-value iteration would silently produce empty mappings, making diagnosis non-trivial and requiring approximately four days of debugging. The resolution required a custom string-manipulation parser:

```
video_id = url.split('/')[−1].replace('.mp4','')
```

A concurrent analysis revealed extreme class imbalances: 297 video clips for ‘Book,’ 1 for ‘Whistle,’ requiring an additional two weeks of dataset curation.

C. Failure Point 2: The Skeleton Scout and Undocumented Node Layout

Skeletal extraction targeting documented hand landmark indices 501–543 produced geometrically implausible structures. Diagnosis required building a custom visual inspection tool—the *Skeleton Scout*—which revealed that the WLASL dataset used an undocumented, highly compressed 180-node layout entirely different from the standard MediaPipe schema. The complete undocumented layout, as mapped by the Skeleton Scout, is shown in Table I.

TABLE I
UNDOCUMENTED 180-NODE WLASL SKELETON LAYOUT AS MAPPED
BY THE SKELETON SCOUT

Node Range	Anatomical Region	Count	Decision
0–41	Hands (21×2 hands)	42	RETAINED
42–47	Arms and Shoulders	6	DISCARDED
48–179	Face, Head, and Body	132	DISCARDED

TABLE II
MEDIAPIPE-LSTM PROTOTYPE BENCHMARK RESULTS—RADXA ZERO
3W (ARM CORTEX-A53)

Measurement	Result	Threshold	Status
LSTM FP32 Inference Latency	47 ms	≤ 33 ms	✗ Fail
LSTM INT8 Inference Latency	38 ms	≤ 33 ms	✗ Fail
LSTM FP32 Accuracy	89.4%	$> 90\%$	✗ Fail
LSTM INT8 Accuracy	85.2%	$> 90\%$	✗ Fail
INT8 Accuracy Degradation	4.2%	$< 1\%$	✗ Fail
Data Engineering Duration	~ 6 wks	—	High

D. Failure Point 3: Real-Time Constraint Violation Under INT8 Quantization

With correct skeletal geometry, the trained LSTM achieved 89.4% top-1 accuracy on the WLASL held-out test set in FP32 precision. Hardware benchmarking on the Radxa Zero 3W Cortex-A53 produced the disqualifying results shown in Table II.

The 4.2% accuracy drop under INT8 quantization is attributable to the sensitivity of LSTM gate activation functions (sigmoid and tanh) to quantization noise. Unlike ReLU activations used in CNNs—which are linear for positive inputs and thus well-behaved under affine INT8 approximation—sigmoid and tanh are smooth nonlinear functions whose curvature introduces quantization error that accumulates across the hidden state recurrence.

Pivot Decision. The MediaPipe-LSTM skeleton prototype was formally discontinued after 6 weeks of engineering effort. The pivot to ResNet18 direct image classification was motivated by three simultaneous disqualifying failures: LSTM inference exceeds the real-time budget even after quantization; INT8 quantization of LSTM gates causes a 4.2% accuracy drop—four times the acceptable threshold; and the skeleton extraction pipeline requires weeks of forensic engineering for undocumented dataset anomalies.

V. THE VOXERA DATA ENGINEERING PIPELINE

A. Data Collection: The POV-ISL Dataset

All recording was performed using the OmniVision OV5647 5-megapixel fisheye camera module mounted on the Voxera wearable harness, connected via MIPI CSI-2 to the Radxa Zero 3W. Video was captured at 640×480 pixels at 30 fps in YUV420 colour space. Forty ISL sign classes were recorded: the 26 letters of the ISL manual alphabet (A–Z) and 14 common static-space vocabulary words (*Yes, No, Hello, Thank You, Help, Please, Name, Where, How, What, I, You, Water,*

Food). Multiple signers of varied body sizes, arm lengths, skin tones, and clothing were recorded across three distinct environments.

B. Frame Extraction

For each recorded video clip, 30 frames were extracted using OpenCV’s VideoCapture interface, sampled at uniform intervals across the clip duration, yielding a raw corpus of approximately 12 000 labelled images across 40 classes.

C. Aggressive Offline Image Augmentation

A raw dataset of 300 images per class is grossly insufficient to fine-tune an 11.7-million-parameter ResNet18 without severe overfitting. We implemented an aggressive offline augmentation pipeline expanding the corpus $8\times$ to approximately 96 000 training images. All augmentation was applied exclusively to the training partition; the validation partition uses raw, unaugmented images only.

1) *Randomized Brightness Shifting*: A multiplicative brightness factor β is sampled uniformly from $[0.7, 1.3]$. Applied to the V channel of the image after HSV conversion:

$$V'(x, y) = \text{clip}(\beta \cdot V(x, y), 0, 255). \quad (1)$$

2) *Gaussian Noise Injection*: Zero-mean Gaussian noise is added independently to each pixel and channel:

$$P'(x, y, c) = \text{clip}\left(P(x, y, c) + \mathcal{N}(0, \sigma^2), 0, 255\right), \quad (2)$$

where σ is sampled uniformly from $[0.01 \times 255, 0.05 \times 255]$.

3) *2D Affine Rotation*: A rotation angle θ is sampled uniformly from $[-15^\circ, +15^\circ]$. The image is rotated about its centre (c_x, c_y) :

$$x' = (x - c_x) \cos \theta - (y - c_y) \sin \theta + c_x, \quad (3)$$

$$y' = (x - c_x) \sin \theta + (y - c_y) \cos \theta + c_y. \quad (4)$$

4) *Horizontal Flipping*: The image is horizontally mirrored with 50% probability, doubling the effective training distribution while maintaining label correctness for bilaterally symmetric handshapes.

5) *Random Crop-and-Resize*: A random crop region of dimensions $(r_H \times H, r_W \times W)$ is selected, where r is sampled uniformly from $[0.90, 1.00]$, then resized to the standard 224×224 input resolution via bilinear interpolation.

6) *Colour Jitter*: The image hue is shifted by $\Delta h \sim \mathcal{U}(-18^\circ, +18^\circ)$ and saturation is scaled by $\Delta s \sim \mathcal{U}(0.8, 1.2)$:

$$H' = (H + \Delta h) \bmod 180, \quad S' = \text{clip}(S \times \Delta s, 0, 255). \quad (5)$$

Table III summarises all augmentation operations and their physical interpretation. Table IV shows the dataset partitioning statistics.

TABLE III
AUGMENTATION OPERATIONS—PARAMETERS AND PHYSICAL INTERPRETATION

Augmentation	Parameters	Phenomenon Simulated
Brightness Shift	$\beta \in [0.70, 1.30]$	Varying ambient illumination
Gaussian Noise	$\sigma \in [2.6, 12.8]$	Sensor thermal/electronic noise
Affine Rotation	$\theta \in [-15^\circ, +15^\circ]$	Wrist tilt and posture variation
H-Flip	$p = 0.50$	Lateral perspective variation
Crop & Resize	$r \in [0.90, 1.00]$	Signer-camera distance variation
Colour Jitter	$\Delta h \in \pm 18^\circ$, $\Delta s \in [0.8, 1.2]$	Diverse skin tones and backgrounds

TABLE IV
DATASET PARTITIONING AND AUGMENTATION STATISTICS

Partition	Raw Images	After 8×	Usage
Training (80%)	~9 600	~76 800	Weight updates
Validation (20%)	~2 400	~2 400	Hyperparameter selection
Test (held-out)	~1 200	~1 200	Final evaluation
Total	~13 200	~80 400	—

VI. THE AI ARCHITECTURE: RESNET18 WITH CUSTOM CLASSIFICATION HEAD

A. Architecture Selection: Multi-Criteria Analysis

The core classification problem—mapping a single 224×224 RGB frame to one of $C = 40$ ISL sign class labels—requires a convolutional architecture satisfying four simultaneous criteria: (1) accuracy $>90\%$ on the 40-class ISL vocabulary; (2) INT8 quantization accuracy degradation $<1\%$; (3) INT8 inference latency ≤ 33 ms on the Cortex-A53; and (4) compatibility with ImageNet transfer learning. Table V shows the three candidate architectures benchmarked on the target hardware.

MobileNetV2 was rejected due to its accuracy of 89.2% falling below the 90% threshold and INT8 degradation of 3.1% exceeding the 1% limit. ResNet18 was selected as the optimal trade-off across all four criteria.

B. The Residual Block: Mathematical Foundation

ResNet18 is a member of the Residual Network family introduced by He et al. [2], which resolved the degradation problem observed in very deep vanilla CNNs. The residual block’s defining equation is:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x}, \quad (6)$$

where $\mathbf{x} \in \mathbb{R}^{C \times H \times W}$ is the block input tensor, $\mathcal{F}(\mathbf{x}, \{W_i\})$ is the residual function, and $\mathbf{y}$ is the block output before the final ReLU activation. When channel depth or spatial resolution differs between input and output, a 1×1 projection convolution is applied:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + W_s \mathbf{x}. \quad (7)$$

TABLE V
ARCHITECTURE COMPARISON—ALL CANDIDATES EVALUATED ON RADXA ZERO 3W (CORTX-A53)

Architecture	Params	FP32	INT8	Acc. (INT8)	Sel.
MobileNetV2 [5]	3.4 M	28 ms	19 ms	89.2%	No
EfficientNet-Lite0	4.7 M	35 ms	24 ms	91.9%	No
ResNet18 [2]	11.7 M	68 ms	25 ms	91.8%	✓

TABLE VI
RESNET18 LAYER-BY-LAYER ARCHITECTURE SUMMARY WITH CUSTOM CLASSIFICATION HEAD

Stage	Blocks	Filters	Spatial Out	Params
Stem	$1 \times \text{Conv}(7 \times 7) + \text{MaxPool}$	64	56×56	9 408
Stage 1	$2 \times \text{BasicBlock}$	64	56×56	147 968
Stage 2	$2 \times \text{BasicBlock (stride-2)}$	128	28×28	525 568
Stage 3	$2 \times \text{BasicBlock (stride-2)}$	256	14×14	2 099 200
Stage 4	$2 \times \text{BasicBlock (stride-2)}$	512	7×7	8 393 728
GAP + Head	—	—	40	131 368
Total	—	—	—	11 307 240

During backpropagation, the gradient of the loss $\mathcal{L}$ with respect to the block input $\mathbf{x}$ is:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}} = \frac{\partial \mathcal{L}}{\partial \mathbf{y}} \cdot \left(\frac{\partial \mathcal{F}}{\partial \mathbf{x}} + \mathbf{I} \right). \quad (8)$$

The identity term $\mathbf{I}$ guarantees that gradient signal flows directly from any later layer back to any earlier layer, eliminating the vanishing gradient problem and enabling effective domain adaptation of the full feature hierarchy. Table VI summarises the layer-by-layer architecture.

C. Transfer Learning Strategy: Two-Phase Fine-Tuning

The ResNet18 backbone was initialized with weights pre-trained on ImageNet. A two-phase fine-tuning strategy was employed. In Phase 1 (epochs 1–30), the backbone weights were frozen and only the custom classification head was trained. In Phase 2 (epochs 31–60), all weights were unfrozen and the full network was trained end-to-end, with the backbone learning rate set one order of magnitude below the head learning rate (10^{-5} vs. 10^{-4}).

D. Custom Classification Head

The original ResNet18 classification head was replaced with a custom head for the 40-class ISL vocabulary:

- **Linear**(512 $\rightarrow$ 256): Reduces the 512-dimensional ResNet18 feature embedding to a 256-dimensional ISL-specific representation. Trainable parameters: $512 \times 256 + 256 = 131\,328$.
- **ReLU Activation**: Introduces non-linearity for learning non-linear classification boundaries.
- **Dropout (rate = 0.40)**: Randomly zeroes 40% of the 256 embedding dimensions during training. Rate determined via ablation (Section VIII).
- **Linear**(256 $\rightarrow$ 40): Maps the regularized embedding to 40 raw logit scores. Trainable parameters: $256 \times 40 + 40 = 10\,280$.

TABLE VII
TRAINING HYPERPARAMETER CONFIGURATION—TWO-PHASE
FINE-TUNING

Hyperparameter	Phase 1 (1–30)	Phase 2 (31–60)
Backbone Weights	Frozen	Trainable
Head Learning Rate	1×10^{-4}	1×10^{-4}
Backbone LR	N/A	1×10^{-5}
LR Decay	$\times 0.5 / 10$ ep plateau	$\times 0.5 / 10$ ep plateau
Batch Size	32	32
Dropout Rate	0.40	0.40
Loss	Cross-Entropy	Cross-Entropy
Optimizer	Adam	Adam

- **Softmax:** Converts the 40 logits $\mathbf{z} = [z_1, z_2, \dots, z_{40}]$ to a probability distribution:

$$P(\text{class } i \mid \mathbf{x}) = \frac{\exp(z_i)}{\sum_{j=1}^{40} \exp(z_j)}. \quad (9)$$

E. Training Configuration and Loss Function

Training was performed using the Adam optimizer [8] with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 1 \times 10^{-8}$. The loss function is categorical cross-entropy:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^{40} y_{ij} \log P(\text{class } j \mid \mathbf{x}_i), \quad (10)$$

where y_{ij} is the one-hot ground-truth label for sample i , class j . A plateau-based learning rate decay was applied: if the validation loss did not decrease for 10 consecutive epochs, the learning rate was multiplied by 0.5. Table VII shows the full hyperparameter configuration.

F. INT8 Post-Training Quantization via TFLite

Following training, the PyTorch model was exported to ONNX and converted to TensorFlow Lite [6] with INT8 post-training quantization. The quantization process maps each FP32 tensor value x to an INT8 integer x_q through a per-layer affine mapping:

$$x_q = \text{clamp}\left(\left\lfloor \frac{x}{\text{scale}} \right\rfloor + \text{zero_point}, -128, 127\right), \quad (11)$$

where per-layer scale and zero_point are determined during a calibration step using 200 representative ISL images:

$$\text{scale} = \frac{x_{\max} - x_{\min}}{2^8 - 1}, \quad \text{zero_point} = \text{clamp}\left(\left\lfloor -\frac{x_{\min}}{\text{scale}} \right\rfloor, -128, 127\right), \quad (12)$$

Per-layer calibration is essential because activation ranges vary substantially across ResNet18 stages (Stage 1: ~ 0 –3; Stage 4: ~ 0 –12), explaining the low 0.5% accuracy cost of quantization for ResNet18 versus the 4.2% cost for LSTM.

VII. HARDWARE DEPLOYMENT—THE RADXA ZERO 3W EDGE PLATFORM

A. Compute Platform

The Radxa Zero 3W is an ultra-compact single-board computer (65 mm $\times$ 37 mm) powered by the Rockchip RK3566 System-on-Chip. The RK3566 integrates a quad-core ARM

TABLE VIII
RADXA ZERO 3W HARDWARE SPECIFICATIONS

Component	Specification	Relevance
SoC	Rockchip RK3566	CPU + GPU + ISP
CPU	4 $\times$ Cortex-A53 @ 1.8 GHz	NN inference host
SIMD	128-bit NEON (16 INT8 MACs/cycle)	ResNet18 INT8 Conv2D
GPU	ARM Mali-G52	Camera ISP only
RAM	2 GB LPDDR4 @ 3200 MT/s	Model + frame buffers
Storage	32 GB eMMC	Model, OS, data
Camera I/F	MIPI CSI-2 (1-lane)	DMA frame capture
Audio I/F	I2S	Speaker output (TTS)
Wireless	Wi-Fi 5 + Bluetooth 5.0	OTA + smartwatch
Peak Power	3.2 W	Battery sustainable
Battery Life	~ 6.5 h (5 000 mAh)	Full-day operation
Form Factor	65 $\times$ 37 $\times$ 8 mm	Wearable harness
BOM Cost	$\sim$ USD 85	<2 days interpreter fees

Cortex-A53 processor running at up to 1.8 GHz with ARMv8-A 64-bit architecture, a Mali-G52 GPU (used for camera ISP offloading, not for neural network inference), 2 GB of LPDDR4 RAM at 3200 MT/s, and a dedicated MIPI CSI-2 camera interface. The Cortex-A53 NEON SIMD unit supports 16 simultaneous INT8 multiply-accumulate operations per clock cycle, directly accelerating the INT8-quantized convolutional kernels. Table VIII shows full hardware specifications.

B. Camera and Sensor Integration

The OmniVision OV5647 5-megapixel fisheye camera module is connected via MIPI CSI-2, providing a dedicated DMA-driven frame transfer pathway. The camera is configured at 640 $\times$ 480 pixels at 30 fps in YUV420 colour space. The YUV420 format reduces MIPI bus bandwidth to 1.5 bytes/pixel (vs. 3 bytes/pixel for RGB888), enabling reliable 30 fps transfer over the single-lane CSI-2 interface. Conversion from YUV420 to RGB888 and resizing from 640 $\times$ 480 to 224 $\times$ 224 are performed on-CPU as the first stage of the inference pipeline, consuming approximately 3 ms.

C. Audio Output and Text-to-Speech Pipeline

The primary communication modality of Voxera is spoken audio. The audio pipeline is: recognized sign label $\rightarrow$ pyttsx3 TTS synthesis using espeak-ng backend $\rightarrow$ PCM audio waveform $\rightarrow$ RK3566 I2S interface $\rightarrow$ 1 W Class-D amplifier $\rightarrow$ 8-ohm miniature speaker. The espeak-ng engine operates entirely on-device without internet connectivity, consuming approximately 4 ms per word on the Cortex-A53. A secondary Bluetooth Low Energy channel transmits UTF-8 text to a paired smartwatch for silent visual communication.

D. End-to-End Latency Decomposition

Table IX shows the complete latency decomposition for the sign-to-speech pipeline.

TABLE IX
END-TO-END SYSTEM LATENCY DECOMPOSITION

Pipeline Stage	Latency	Notes
Frame capture (MIPI CSI-2, DMA)	33 ms	Overlapped with prior inference
YUV420→RGB888 + resize	~3 ms	Sequential after frame ready
Mean subtraction (normalise)	<1 ms	Sequential
ResNet18 INT8 TFLite inference	~25 ms	NEON SIMD parallelised
Argmax + vocabulary lookup	<1 ms	Sequential
espeak-ng TTS synthesis	~4 ms/word	Single-threaded
I2S audio + speaker	~21 ms	Acoustic propagation
Total	87 ms ± 12 ms	Below 150 ms perceptual threshold

VIII. EXPERIMENTAL RESULTS AND ABLATION STUDY

A. Classification Performance—Primary Results

Table X shows the primary classification results. The 0.5% accuracy degradation from FP32 to INT8 quantization falls within the 95% confidence interval ($\pm 0.9\%$) of the evaluation, indicating that the quantization accuracy cost is not statistically distinguishable from evaluation noise.

B. Architecture Comparison: MediaPipe-LSTM vs. ResNet18

Table XI provides the full architecture comparison between the discontinued prototype and the deployed system.

C. Ablation Study

Table XII quantifies the contribution of individual design decisions to final performance. Three dominant contributions are identified: (1) the augmentation pipeline overall (-13.5% without augmentation); (2) ImageNet pre-training (-8.2% without); and (3) dropout regularization (-3.7% without).

D. Per-Environment Background Invariance Evaluation

Table XIII shows the per-environment accuracy assessment. The 4.1 percentage-point accuracy spread across environments (89.7% to 93.8%) represents a substantial improvement over the non-augmented baseline (17.8 percentage-point spread), demonstrating that the augmentation strategy reduces environment-induced accuracy variance by approximately 77%.

E. Per-Class Analysis and Primary Confusion Patterns

Per-class F1-scores across the 40 ISL classes range from 88.1% to 97.3%. The five lowest-performing classes and their primary confusion partners are shown in Table XIV. These confusion patterns are consistent with published ISL and ASL alphabet recognition literature: the highest-confusion pairs involve handshapes that differ only in the position of a single finger or the thumb.

TABLE X
PRIMARY CLASSIFICATION RESULTS—RESNET18 FP32 VS. INT8
(40-CLASS ISL)

Metric	FP32	INT8	Δ
Top-1 Accuracy	92.3%	91.8%	-0.5%
Top-5 Accuracy	99.1%	98.7%	-0.4%
Macro-Avg Precision	91.8%	91.4%	-0.4%
Macro-Avg Recall	92.0%	91.6%	-0.4%
Macro-Avg F1-Score	91.9%	91.5%	-0.4%
Inference Latency	68 ms	25 ms	-63%
Model Size	44.7 MB	11.2 MB	-75%
95% CI of Accuracy	$\pm 0.8\%$	$\pm 0.9\%$	—

TABLE XI
FULL ARCHITECTURE COMPARISON—DISCONTINUED PROTOTYPE VS.
RESNET18 SYSTEM

Metric	LSTM (disc.)	ResNet18 INT8
Test Accuracy	85.2% (INT8)	91.8%
INT8 Quant. Drop	4.2%	0.5%
Cortex-A53 Latency	38 ms (INT8)	25 ms
Meets 33 ms Budget	No ✗	Yes ✓
Data Eng. Duration	~6 wks	~2 wks
Dataset Source	WLASL (exocentric)	POV-ISL (egocentric)
Domain Shift Risk	High (~18.4%)	Eliminated
Pipeline Fail Points	6	3

TABLE XII
ABLATION STUDY—CONTRIBUTION OF INDIVIDUAL SYSTEM COMPONENTS (INT8, 40-CLASS ISL)

Ablation Condition	Acc. (INT8)	Δ
Full System (baseline)	91.8%	—
No augmentation (raw data only)	78.3%	−13.5%
Brightness aug. removed	88.7%	−3.1%
Gaussian noise aug. removed	90.1%	−1.7%
H-flip aug. removed	90.9%	−0.9%
Colour jitter removed	89.3%	−2.5%
Random crop removed	90.6%	−1.2%
Affine rotation removed	91.0%	−0.8%
Phase 1 training skipped (direct E2E)	89.4%	−2.4%
Dropout 0.0 (no regularization)	88.1%	−3.7%
Dropout 0.20 (lower)	90.7%	−1.1%
Dropout 0.40 (selected)	91.8%	—
Dropout 0.60 (higher)	90.2%	−1.6%
MobileNetV2 backbone	89.2%	−2.6%
Random ImageNet init (no pretraining)	83.6%	−8.2%

TABLE XIII
PER-ENVIRONMENT ACCURACY ASSESSMENT—BACKGROUND INVARIANCE QUANTIFICATION

Environment	Images	Top-1 (INT8)
Indoor office (fluorescent)	400	93.8%
Outdoor daylight (variable)	400	90.4%
Low-light indoor (<50 lux)	400	89.7%
All environments combined	1 200	91.8%
Without augmentation (same)	1 200	82.3%

TABLE XIV
LOWEST-PERFORMING ISL CLASSES AND PRIMARY CONFUSION PARTNERS

Class	F1	Distinguishing Feature
‘M’	88.1%	Fingers folded over thumb (3 vs. 2 for ‘N’)
‘S’	88.9%	Thumb position relative to folded fingers
‘U’	89.4%	Degree of finger spread vs. ‘V’
‘R’	89.7%	Finger crossing—small spatial overlap
‘T’	90.1%	Thumb position between vs. over first finger

IX. SOCIETAL IMPACT: ALIGNMENT WITH UN SUSTAINABLE DEVELOPMENT GOALS

A. SDG 4: Quality Education

UN SDG 4 calls for inclusive and equitable quality education for all. For the estimated 1.8 million Deaf school-age children in India, the majority enrolled in mainstream schools without interpreter support, the current reality falls sharply short of this aspiration [1]. Project Voxera directly addresses this gap. A Deaf student wearing Voxera can sign a question at any moment; the device produces spoken audio within 87 ms, enabling the teacher to respond in real time. The student reads the teacher’s verbal response on a paired smartwatch display, creating a bilateral, low-latency, setup-free communication loop that makes spontaneous classroom participation structurally equivalent for Deaf and hearing students for the

first time.

B. SDG 8: Decent Work

SDG 8 calls for sustained, inclusive, and sustainable economic growth and decent work for all. For the Deaf professional community, this goal is blocked by what we term the *Glass Ceiling of Communication*: a structurally embedded set of barriers manifesting in spontaneous workplace communication, client-facing roles, leadership positions, and international collaboration [1]. Voxera addresses all four scenarios. At USD 100/hour for interpreter services, a Deaf professional requiring 4 interpreted hours per work week incurs USD 20 800 per year in interpreter costs [1]. Voxera’s USD 85 hardware cost represents a payback period of less than two working days.

C. Ethical Obligations and Community Engagement Commitment

The development of AI-based assistive technology for marginalized linguistic communities carries specific ethical obligations. Voxera Brain 3, developed within an academic project timeframe, did not include formal co-design engagement with the ISL community. Future iterations will incorporate: (1) co-design workshops with ISL-using Deaf adults; (2) formal user studies evaluating real-world performance across diverse signing styles, ages, and regional ISL variants; and (3) publication of the POV-ISL dataset for community validation, critique, and extension.

X. CONCLUSION

This paper has presented Project Voxera: a wearable, egocentric, edge-deployed Indian Sign Language recognition system achieving 91.8% top-1 classification accuracy over a 40-class ISL vocabulary at 25 ms inference latency, enabling sustained 30 fps operation on a USD 85 embedded device without cloud connectivity, specialist sensor hardware, or any environmental setup.

The contributions of this work are four. First, the construction of an original egocentric POV-ISL dataset recorded

from the deployment hardware itself, eliminating domain shift by construction—a domain shift representing a mean accuracy penalty of approximately 18.4 percentage points for exocentric-to-egocentric transfer [4]. Second, the empirical demonstration—via a formal ablation study—that aggressive offline augmentation (8× expansion via six distinct operations) is the dominant contributor to background-invariant performance, accounting for 13.5 of the 91.8 percentage-point final accuracy. Third, the rigorous hardware benchmarking and formal documentation of the discontinued MediaPipe-LSTM prototype’s three failure modes on the target Cortex-A53 platform, constituting a replicable negative result for the edge-deployed sign recognition literature. Fourth, the demonstration that ResNet18 direct image classification achieves simultaneously higher accuracy, lower quantization degradation, lower inference latency, and lower data engineering complexity than the skeleton-based LSTM architecture on identical hardware.

The transition from a static-image laboratory prototype to a dynamic, production-ready edge product is structured as a four-phase technical roadmap, described below.

Phase 1: Synthetic Data Pipeline via Blender Domain Randomization. Scaling the ISL vocabulary beyond 40 signs without manually recording thousands of hours of video requires automated synthetic data generation. The planned pipeline uses a Python `bpy` script inside Blender to automatically render 3D hand poses at scale. Critically, domain randomization must be applied so that rendered images mimic the imperfections of cheap edge-device cameras rather than appearing photorealistic—otherwise the model will fail to generalize to real-world inputs. The pipeline will programatically inject: (1) lens distortion replicating cheap plastic webcam optics; (2) motion blur appropriate to dynamic sign trajectories; (3) sensor ISO noise and low-light luminance degradation; and (4) aggressive JPEG compression at 70% quality. The outcome is a massive, automatically generated training corpus whose appearance distribution closely matches the OV5647 fisheye camera on the deployed wearable.

Phase 2: Spatiotemporal Modeling for Dynamic Sign Recognition. The current ResNet18 architecture classifies static frames; full ISL requires understanding movement over time. Two candidate architectures are under evaluation. *Option A (Motion History Images, MHI)* is a lightweight approach in which OpenCV frame-subtraction is used to accumulate consecutive difference frames into a single “ghostly trail” image encoding the motion trajectory, which is then classified by the existing ResNet18—requiring no architectural changes. *Option B (Sub-sampled CNN + 1D-LSTM)* is the industry-standard approach: ResNet18 is replaced by a heavily quantized MobileNetV3 backbone; five keyframes are sampled uniformly across a one-second motion window; the per-frame feature embeddings are extracted and passed as a sequence into a compact 1D-LSTM to learn the temporal dynamics of the sign. Option B is the primary target architecture for the production system.

Phase 3: Vocabulary Scaling and Non-Manual Marker Integration. As dynamic modeling is established, the system’s

linguistic coverage will be expanded in two directions. First, the sign vocabulary will be scaled from 40 to 400+ signs using curriculum learning—training on high-confidence, visually distinct signs first and progressively introducing confusable or low-frequency signs—combined with transfer learning from pre-trained gesture foundation models. Second, a secondary facial landmark tracking channel will be added to detect Non-Manual Markers (NMMs): eyebrow position, mouth aperture, and head tilt that grammatically differentiate signs sharing identical hand configurations. This NMM channel is necessary for syntactically unambiguous recognition of questions, negations, and topic–comment constructions in fluent ISL.

Phase 4: NPU Hardware Upgrade and Community Co-Design Validation. Running video-sequence models at 30+ fps imposes compute demands that exceed the Cortex-A53’s throughput budget. The hardware platform will transition from general-purpose ARM CPU inference to dedicated Neural Processing Units: candidates include the Google Coral Edge TPU and Radxa boards integrating the Rockchip RK3588 or RK3576 NPU, both of which support INT8 convolutional inference at substantially higher sustained throughput than NEON SIMD alone. In parallel, formal community co-design partnerships will be established with Deaf adults who use ISL as their primary language. Structured user studies will evaluate real-world recognition performance across regional ISL dialect variation, a range of signing speeds, and diverse signer demographics—providing the external validation necessary to assess genuine field utility beyond controlled laboratory conditions.

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